IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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October 19, 2026



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Methodology

- Collected launch data from the SpaceX REST API and Wikipedia (web scraping)
- Cleaned the data and built a landing label: Class = 1 if the first stage landed
- Explored it with pandas/seaborn charts and SQL (SQLite)
- Built an interactive Folium map and a Plotly Dash dashboard
- Trained and tuned 4 classifiers (LogReg, SVM, Decision Tree, KNN) with 10-fold grid search

Results

90
Falcon 9 launches
2010 – 2020

66.7%
first stages
landed

83.3%
test accuracy,
all 4 models

- Landing success rose from 0% (2010–13) to 84–90% (2019–20)
- KSC LC-39A is the strongest site: 77% success, 41.7% of all successful landings
- SSO, VLEO and LEO missions land most often; GTO is hardest (52%)
- The Decision Tree has the best cross-validation accuracy (87.7%)

Introduction

Background

SpaceX advertises Falcon 9 launches at about $62 million; other providers charge upward of $165 million.

Most of the saving comes from landing and reusing the first stage.

If we can predict whether the first stage will land, we can estimate the cost of a launch — useful for a new company ("Space Y") bidding against SpaceX.

Questions we answer

- Which factors — launch site, payload mass, orbit, booster version, experience — drive a successful landing?
- How have landing outcomes changed over time?
- Where are launch sites located, and why there?
- Can a machine-learning model predict the landing outcome, and which model works best?

Section 1

Methodology



Methodology

Executive Summary

1 Data collection SpaceX REST API (launches, rockets, launchpads, payloads, cores) and web scraping of Wikipedia's Falcon 9 launch list

2 Data wrangling Filtered to Falcon 9, imputed missing payload masses, and turned 8 landing outcomes into a binary Class label

3 EDA Scatter, bar and line charts with seaborn/matplotlib; 10 SQL queries on SQLite

4 Interactive visual analytics Folium map of sites, outcomes and proximities; Plotly Dash dashboard with dropdown and payload slider

5 Predictive analysis Standardised features, 80/20 split, GridSearchCV (cv=10) on LogReg, SVM, Decision Tree and KNN; compared on test accuracy and confusion matrices

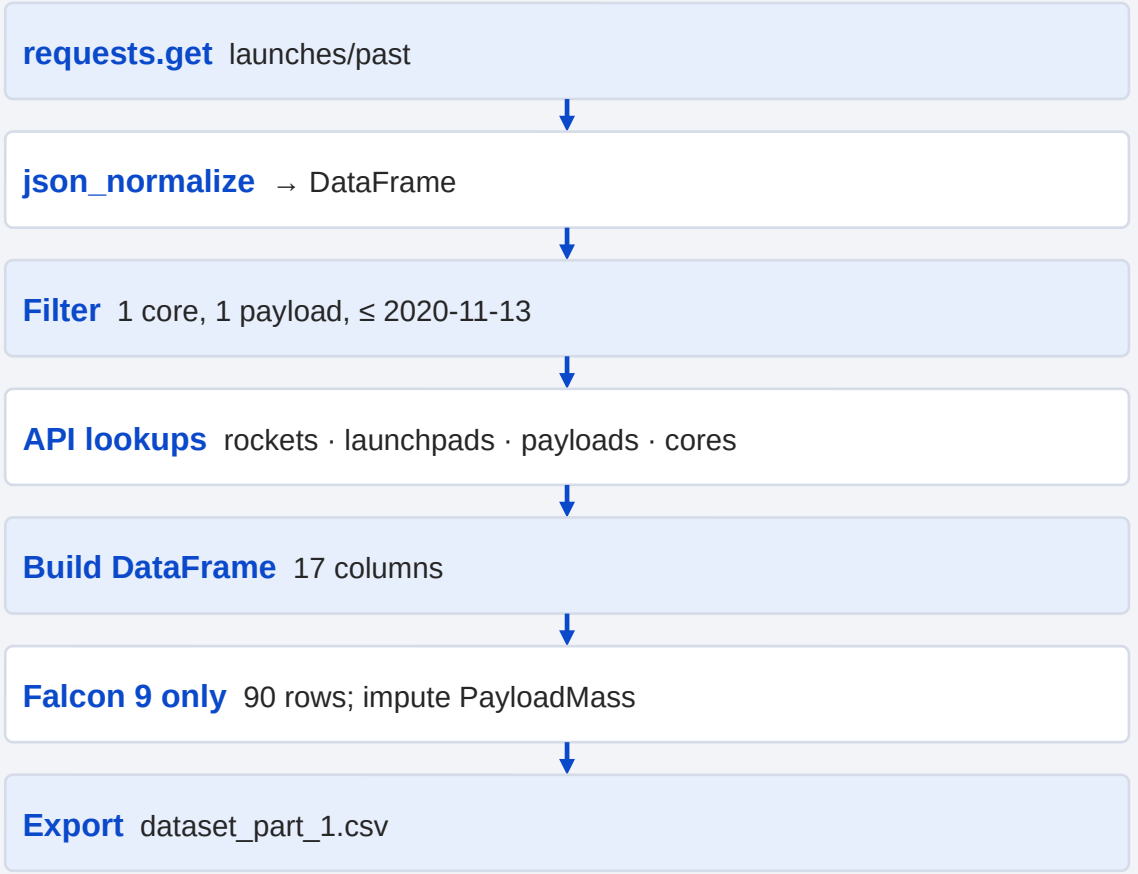
Data Collection

Two public sources were combined into launch-level datasets of Falcon 9 flights:



Data Collection – SpaceX API

- GET /v4/launches/past and decode the JSON with json_normalize
- Keep single-core, single-payload launches up to 13 Nov 2020
- Look up each ID: rockets → booster name; launchpads → site, lat/long; payloads → mass, orbit; cores → outcome, block, reuse, serial
- Keep Falcon 9 only (90 launches) and replace the 5 missing payload masses with the mean (6,105 kg)



GitHub: SpaceX API notebook

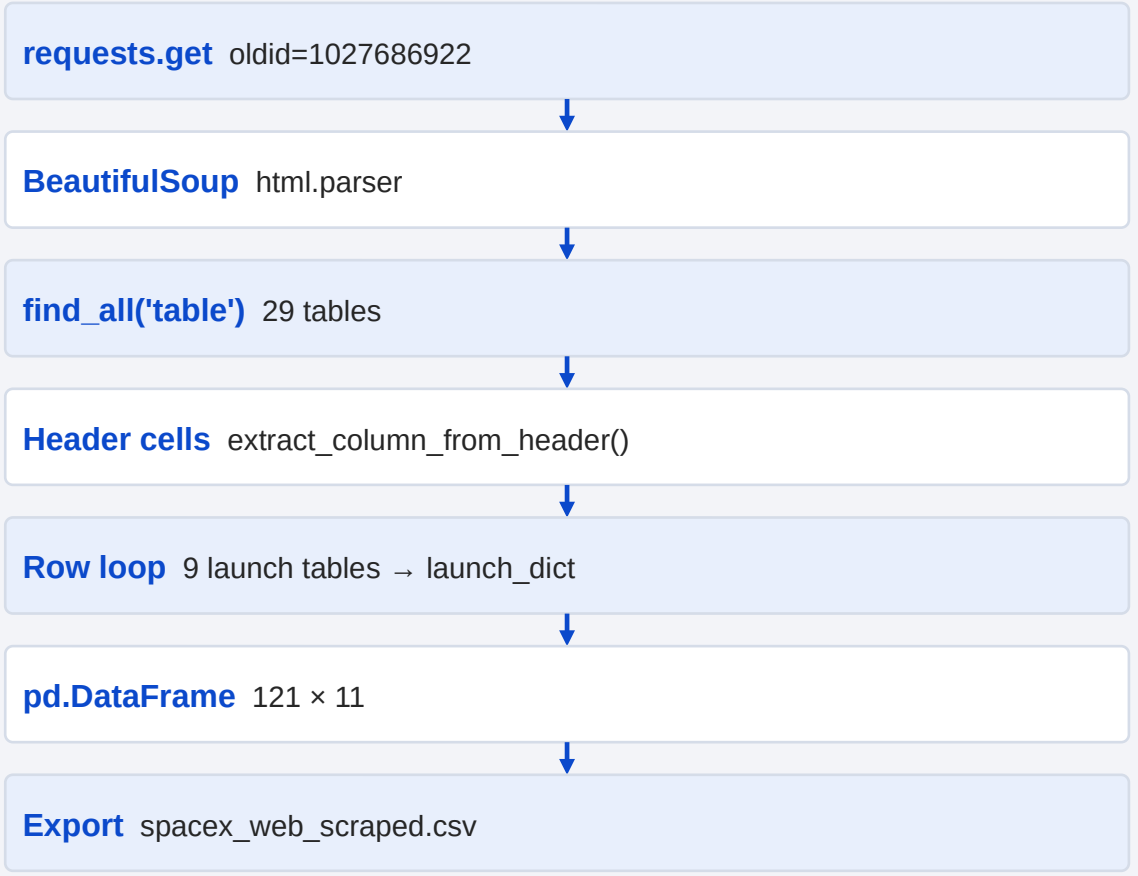
The output matches IBM's reference dataset_part_1.csv exactly.

Data Collection - Scraping

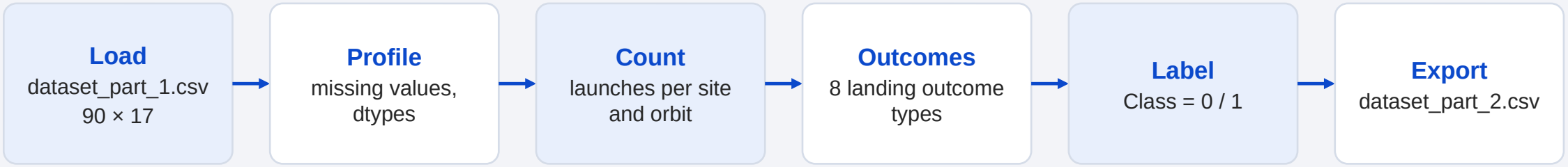
- Request the 9 June 2021 revision of the Wikipedia launch list with requests
- Parse the HTML with BeautifulSoup and collect every <table>
- Read column names from the header cells of the first launch table
- Walk the rows of all launch tables and extract flight no., date, time, booster, site, payload, mass, orbit, customer, launch and landing outcomes

GitHub: Web scraping notebook

Result: 121 launch records, 11 columns.



Data Wrangling



Outcome	Meaning	Count	Class
True ASDS	landed on a drone ship	41	1
True RTLS	landed on a ground pad	14	1
True Ocean	controlled ocean landing	5	1
None None	no landing attempt	19	0
False ASDS	drone-ship landing failed	6	0
False Ocean	ocean landing failed	2	0
None ASDS	drone ship, no attempt	2	0
False RTLS	ground-pad landing failed	1	0

- LandingPad is the only column with gaps (28.9%) — a gap means no pad was used.
- Every outcome that is a failure or no attempt becomes Class 0; landings become Class 1.
- **Overall success rate: 66.7% (60 of 90).**

GitHub: Data wrangling notebook

EDA with Data Visualization

Chart	Type	Why
Flight number vs. launch site	Scatter (catplot)	Does experience at a site change outcomes?
Payload mass vs. launch site	Scatter	Which sites handle heavy payloads, and how well?
Success rate by orbit	Bar chart	Compare success across orbit types
Flight number vs. orbit	Scatter	Does experience matter differently per orbit?
Payload mass vs. orbit	Scatter	Payload/orbit combinations that succeed or fail
Yearly success rate	Line chart	Trend of landing success over time

Feature engineering: selected 12 features and one-hot encoded Orbit, LaunchSite, LandingPad and Serial with get_dummies, then cast everything to float64 (90 × 80) for the models.

GitHub: EDA with data visualization notebook

EDA with SQL

Loaded Spacex.csv into SQLite (SPACEXTABLE) and ran these queries with %sql:

- Unique launch sites (SELECT DISTINCT)
- 5 records for sites beginning with 'CCA' (LIKE, LIMIT)
- Total payload for NASA (CRS) (SUM)
- Average payload of booster F9 v1.1 (AVG)
- Date of the first ground-pad landing (MIN)
- Drone-ship landings with 4,000–6,000 kg payload (AND, ranges)
- Successful vs. failed mission outcomes (GROUP BY, COUNT)
- Boosters that carried the maximum payload (subquery with MAX)
- 2015 drone-ship failures by month (substr)
- Landing outcomes ranked, 2010-06-04 to 2017-03-20 (BETWEEN, ORDER BY)

GitHub: EDA with SQL notebook

Build an Interactive Map with Folium

Object	What it shows	Why
Circle + text label	Each launch site (orange)	See where sites are relative to coasts and the Equator
MarkerCluster	All 56 launches, grouped per site	Many launches share one coordinate; clusters keep the map readable
Marker (green / red)	One per launch, coloured by outcome	Compare success rates between sites at a glance
MousePosition	Lat/long under the cursor	Read coordinates of nearby coastline, rail, roads, towns
PolyLine + distance label	Site → coastline, railway, highway, city	Measure how close sites are to infrastructure and people

Distances were computed with the Haversine formula (calculate_distance) from coordinates read off the map.

GitHub: Folium map notebook

Build a Dashboard with Plotly Dash

Component	Shows / controls	Why
Dropdown (site-dropdown)	All sites or one of 4 sites	Switch between the overall picture and a single site
Pie chart (success-pie-chart)	All sites: share of successes per site One site: success vs. failure	Find the sites with the most and the highest-rate successes
RangeSlider (payload-slider)	Payload 0 – 10,000 kg, 1,000 kg steps	Isolate payload ranges
Scatter (success-payload-scatter-chart)	Payload vs. outcome, coloured by booster version	See how payload and booster version relate to success

Two callbacks connect the inputs to the charts: the dropdown drives the pie chart, and the dropdown plus the slider drive the scatter plot.

GitHub: Plotly Dash app (spacex_dash_app.py)

Predictive Analysis (Classification)



Model	Hyper-parameters searched
Logistic Regression	C ∈ {0.01, 0.1, 1}, L2, lbfgs
Support Vector Machine	kernel (linear, rbf, poly, sigmoid), C and gamma ∈ logspace(-3, 3, 5)
Decision Tree	criterion, splitter, max_depth 2–18, max_features, min_samples_leaf/split
K-Nearest Neighbors	n_neighbors 1–10, algorithm, p ∈ {1, 2}

GitHub: Machine learning prediction notebook

Results

Exploratory analysis

- Success climbs with experience: 25% for flights 1–20 vs. 83% for flights 61–90
- KSC LC-39A and VAFB SLC-4E reach 77%; CCAFS SLC-40 60%
- SSO/VLEO/LEO high, GTO lowest (52%)

Interactive analytics

- All sites sit within ~1–1.5 km of the ocean and ≥ 20 km from towns
- KSC LC-39A: most successes (41.7%) and best ratio (76.9%)
- 2,000–4,000 kg payloads succeed most; FT boosters lead

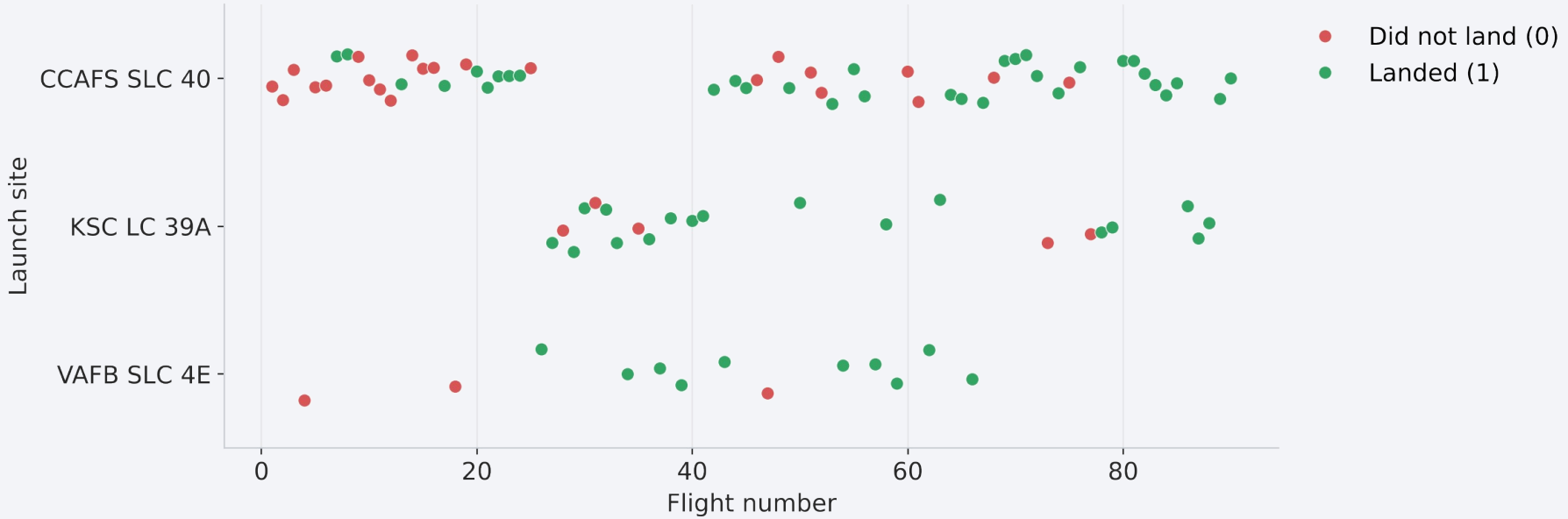
Predictive analysis

- All 4 tuned models: 83.3% on the test set
- Decision Tree best in cross-validation (87.7%)
- Errors are failed landings predicted as landed

Section 2

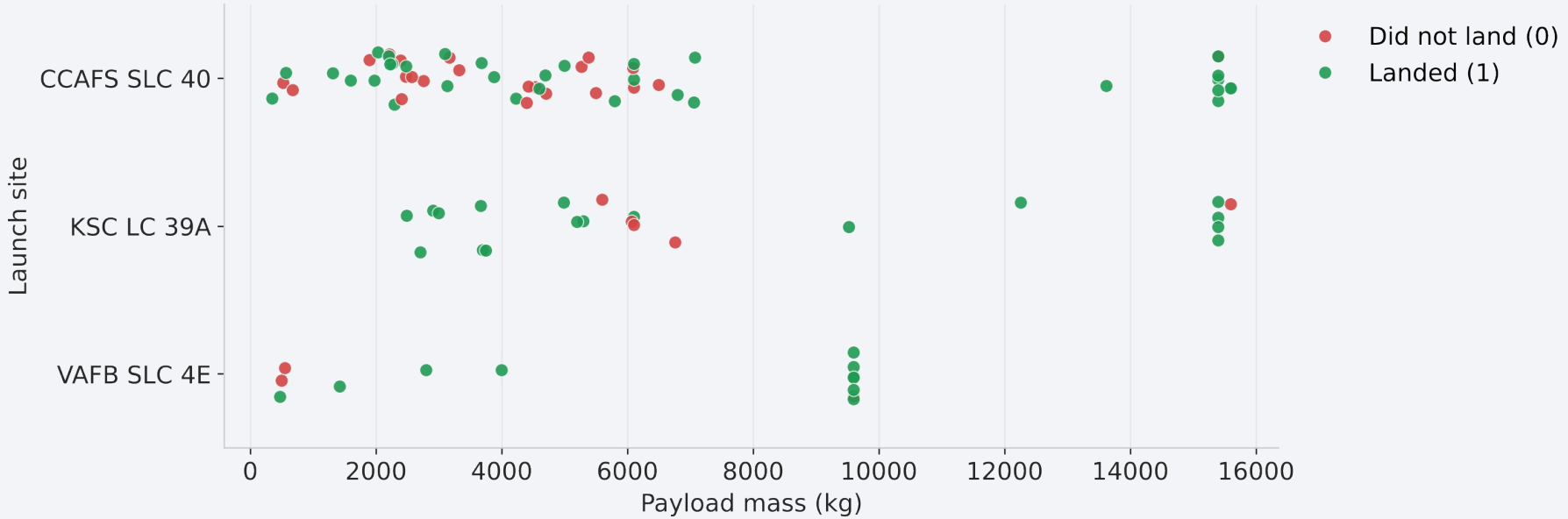
Insights drawn from EDA

Flight Number vs. Launch Site



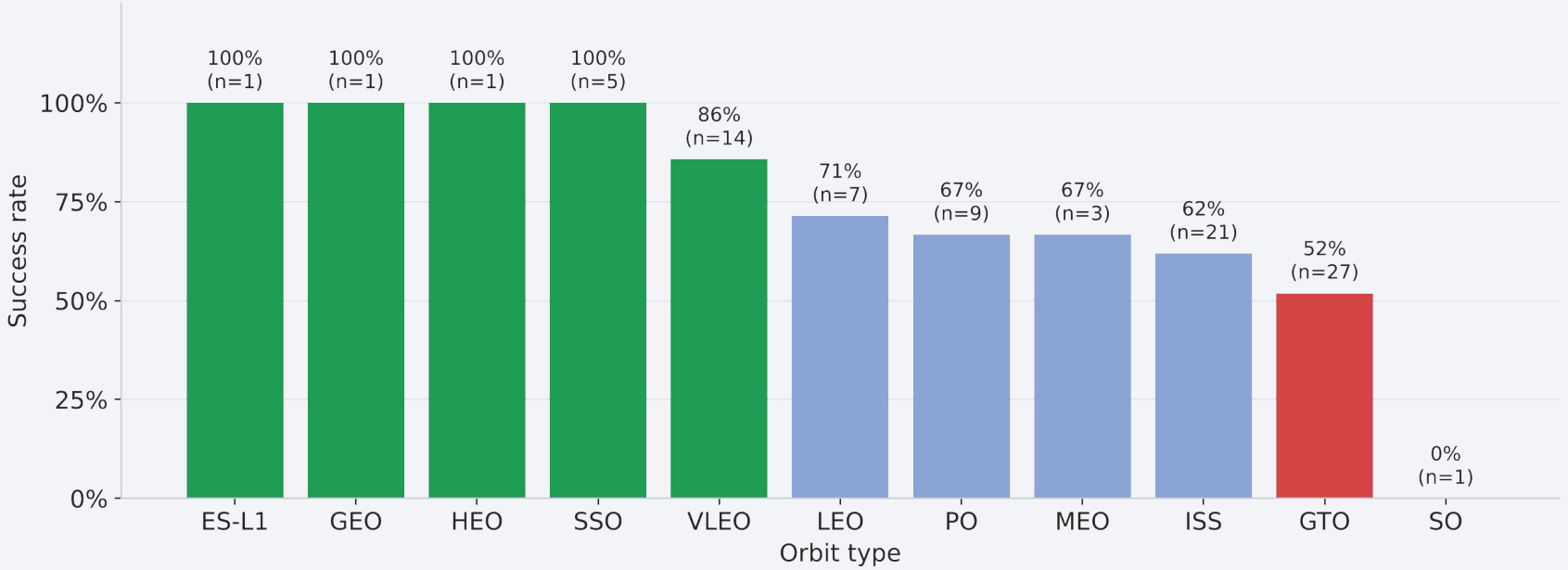
- CCAFS SLC 40 carries most flights (55 of 90), including nearly all early failures.
- Later flights succeed far more often at every site: 25% for flights 1–20 vs. 83% for flights 61–90.
- KSC LC 39A is used from flight 27 onward and has the highest success rate (77%).

Payload vs. Launch Site



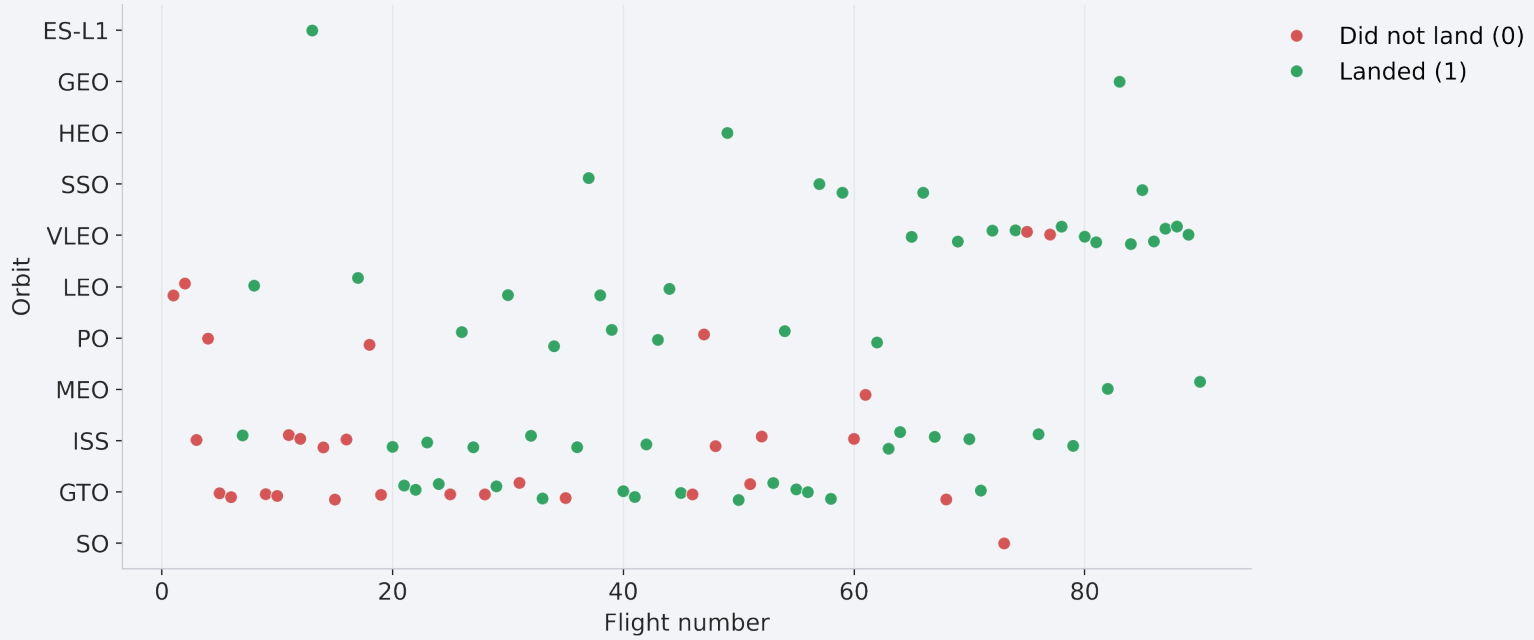
- VAFB SLC 4E never flew more than 9,600 kg; payloads above 10,000 kg flew only from the Florida sites.
- Heavy payloads landed well: 87% success above 8,000 kg vs. 50–70% for lighter ones.
- For light and medium payloads the classes overlap, so payload alone does not decide the outcome.

Success Rate vs. Orbit Type



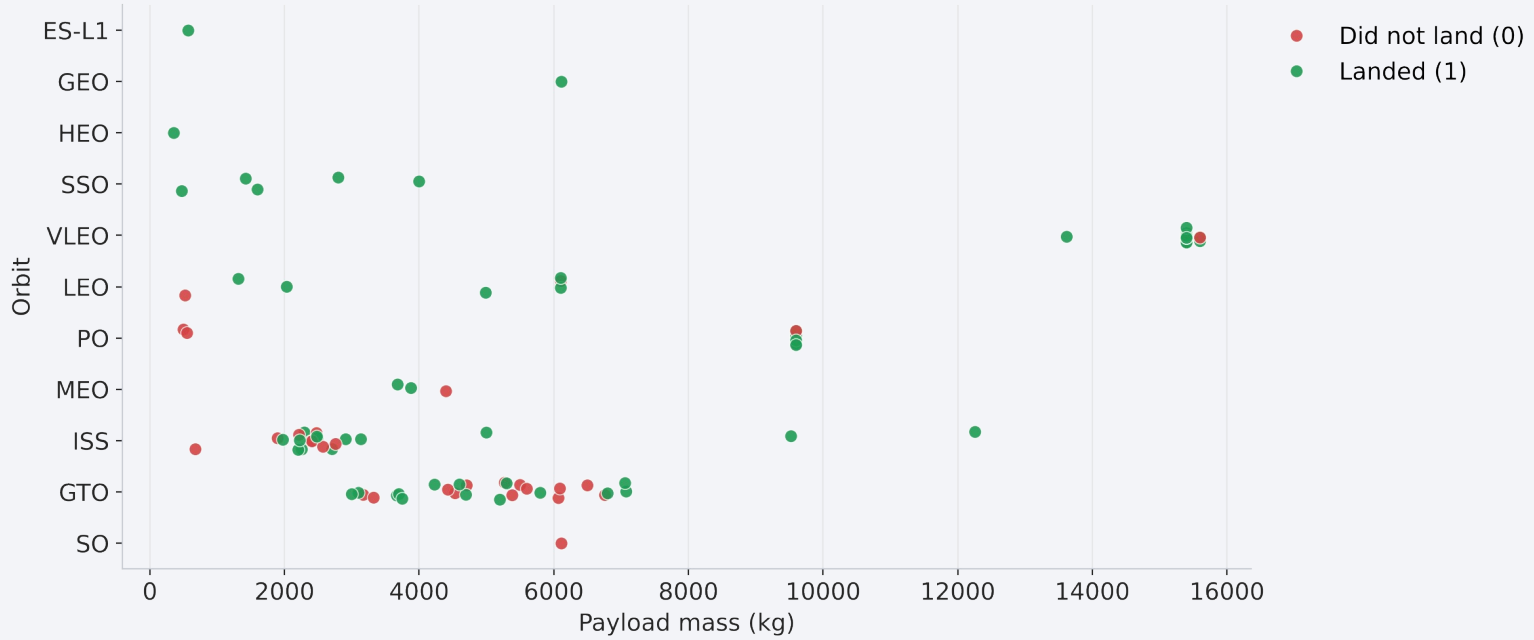
- SSO is the most convincing perfect record (5 of 5); ES-L1, GEO and HEO are 100% but only 1 launch each.
- VLEO (86%, 14 launches) and LEO (71%) are also strong.
- GTO is the hardest common orbit (52% of 27 launches); the single SO launch failed.

Flight Number vs. Orbit Type



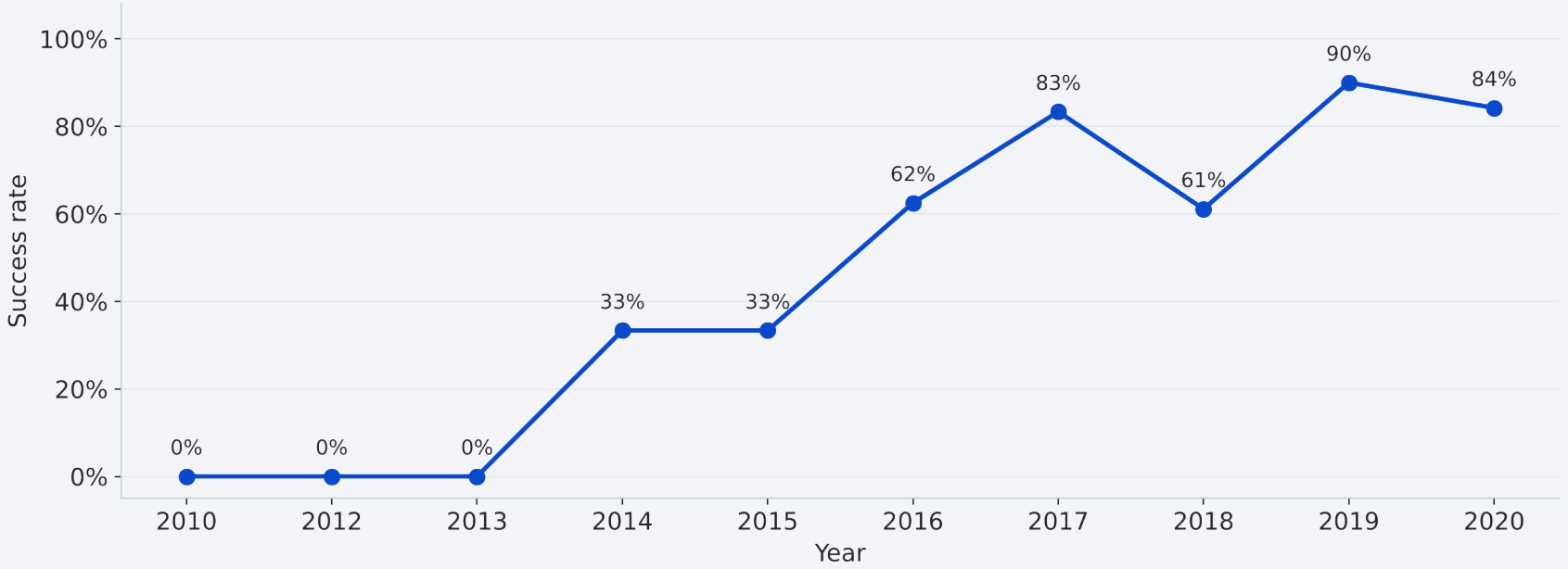
- In LEO, success follows the flight number — early LEO flights failed, later ones landed.
- VLEO launches only start after flight 65 and almost all landed.
- GTO outcomes stay mixed across the whole range of flight numbers.

Payload vs. Orbit Type



- Heavy payloads to Polar, LEO and ISS orbits land successfully more often.
- GTO payloads sit in a narrow 3,000–7,100 kg band where successes and failures overlap.
- VLEO missions (Starlink) carry the heaviest payloads (~15,600 kg) and mostly succeed.

Launch Success Yearly Trend



- No landing succeeded before 2014.
- The success rate climbs from 33% (2014–15) to 62% (2016) and 83% (2017).
- After a dip in 2018 (61%) it reaches 90% in 2019 and 84% in 2020.

All Launch Site Names

```
%sql SELECT DISTINCT "Launch_Site" FROM SPACEXTABLE;
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

SpaceX has used four launch sites: three in Florida (Cape Canaveral and Kennedy Space Center) and one in California (Vandenberg).

DISTINCT removes the repeats across 101 launch records.

Launch Site Names Begin with 'CCA'

```
%sql SELECT * FROM SPACEXTABLE WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5;
```

Date	Time	Booster	Site	Payload	kg	Orbit	Customer	Mission	Landing
2010-06-04	18:45	F9 v1.0 B0003	CCAFS LC-40	Dragon Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

The first five Cape Canaveral launches were early F9 v1.0 Dragon missions to LEO/ISS: every mission succeeded, but no first stage was recovered (parachute failures or no attempt).

Total Payload Mass

```
%sql SELECT SUM("PAYLOAD_MASS__KG_") AS "Total_Payload_Mass_kg"
     FROM SPACEXTABLE WHERE "Customer" = 'NASA (CRS)';
```

Total_Payload_Mass_kg
45,596

Boosters flying NASA's Commercial Resupply Services (CRS) missions carried 45,596 kg in total.

Only rows whose customer is exactly 'NASA (CRS)' are counted.

Average Payload Mass by F9 v1.1

```
%sql SELECT AVG("PAYLOAD_MASS__KG_") AS "Avg_Payload_Mass_kg"
     FROM SPACEXTABLE WHERE "Booster_Version" = 'F9 v1.1';
```

Avg_Payload_Mass_kg
2,928.4

The F9 v1.1 booster carried 2,928.4 kg on average — far less than later Block 5 boosters (up to 15,600 kg).

First Successful Ground Landing Date

```
%sql SELECT MIN("Date") AS "First_Successful_Ground_Pad_Landing"
     FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (ground pad)';
```

First_Successful_Ground_Pad_Landing
2015-12-22

The first successful ground-pad (RTLS) landing happened on 22 December 2015, five and a half years after the first Falcon 9 flight.

Successful Drone Ship Landing with Payload between 4000 and 6000

```
%%sql
SELECT DISTINCT "Booster_Version" FROM SPACEXTABLE
WHERE "Landing_Outcome" = 'Success (drone ship)'
  AND "PAYLOAD_MASS__KG_" > 4000 AND "PAYLOAD_MASS__KG_" < 6000;
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Four Full Thrust (FT) boosters landed on a drone ship with a 4,000–6,000 kg payload.

Two of them (B1021.2, B1031.2) were already re-flown boosters.

Total Number of Successful and Failure Mission Outcomes

```
%%sql
SELECT TRIM("Mission_Outcome") AS "Mission_Outcome", COUNT(*) AS "Total"
FROM SPACEXTABLE GROUP BY TRIM("Mission_Outcome");
```

Mission_Outcome	Total
Failure (in flight)	1
Success	99
Success (payload status unclear)	1

100 of 101 missions succeeded; only one failed in flight.

TRIM merges a 'Success ' value that has a trailing space in the raw data.

Mission success ≠ landing success: the payload reaching orbit says nothing about the first stage.

Boosters Carried Maximum Payload

```
%%sql
SELECT DISTINCT "Booster_Version" FROM SPACEXTABLE
WHERE "PAYLOAD_MASS__KG_" = (SELECT MAX("PAYLOAD_MASS__KG_") FROM SPACEXTABLE);
```

Booster_Version	Booster_Version (cont.)
F9 B5 B1048.4	F9 B5 B1049.5
F9 B5 B1049.4	F9 B5 B1060.2
F9 B5 B1051.3	F9 B5 B1058.3
F9 B5 B1056.4	F9 B5 B1051.6
F9 B5 B1048.5	F9 B5 B1060.3
F9 B5 B1051.4	F9 B5 B1049.7

12 Block 5 boosters carried the maximum payload (15,600 kg — Starlink batches).

All are re-flown boosters (suffix .2 to .7), showing how far reuse had come by 2020.

2015 Launch Records

```
%%sql
SELECT substr("Date", 6, 2) AS "Month", "Landing_Outcome", "Booster_Version", "Launch_Site"
FROM SPACEXTABLE
WHERE substr("Date", 0, 5) = '2015' AND "Landing_Outcome" = 'Failure (drone ship)';
```

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Both 2015 drone-ship failures (January and April) were F9 v1.1 boosters launched from CCAFS LC-40 — the first attempts at landing on a ship.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
%%sql
SELECT "Landing_Outcome", COUNT(*) AS "Outcome_Count" FROM SPACEXTABLE
WHERE "Date" BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY "Landing_Outcome" ORDER BY "Outcome_Count" DESC;
```

Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

In this early period 'No attempt' was most common (10).

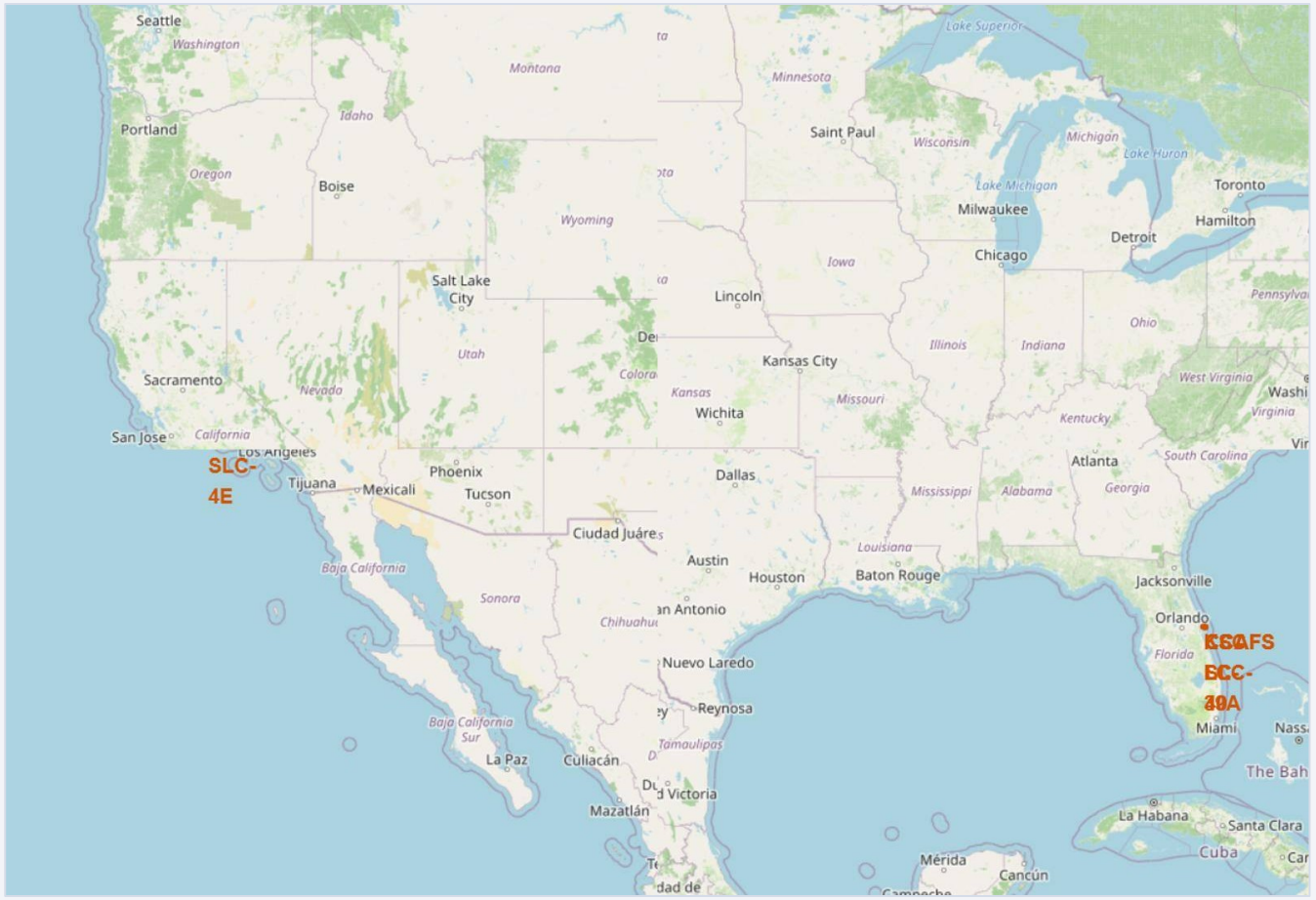
Drone-ship landings were a coin flip (5 successes, 5 failures) and there were 3 ground-pad successes — 8 successful landings out of 31 launches.



Section 3

Launch Sites Proximities Analysis

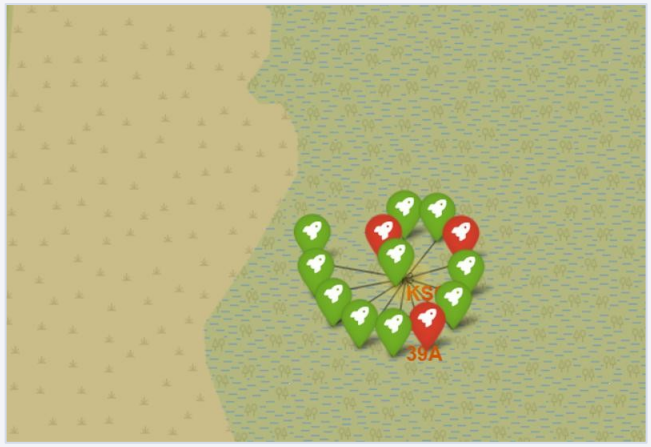
All Launch Sites on a Map

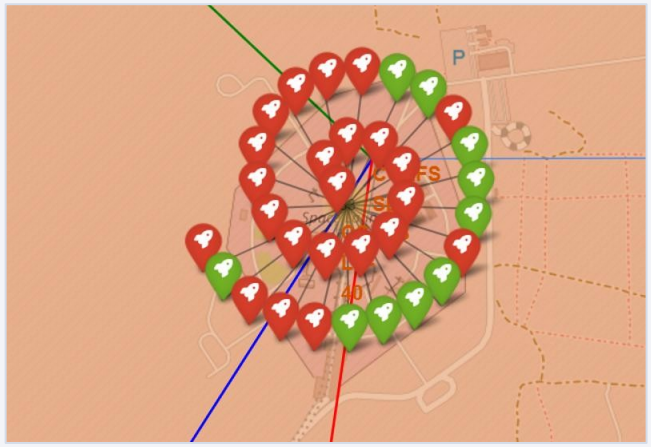


- Orange circles mark the four launch sites: VAFB SLC-4E in California and CCAFS LC-40, CCAFS SLC-40 and KSC LC-39A on Florida's Cape Canaveral.
- No site is near the Equator, but the Florida sites (~28.5° N) are the most southerly options in the continental US, which helps eastward launches.
- Every site sits on the coast, so rockets fly out over the ocean.

Map data © OpenStreetMap contributors

Launch Outcomes by Site

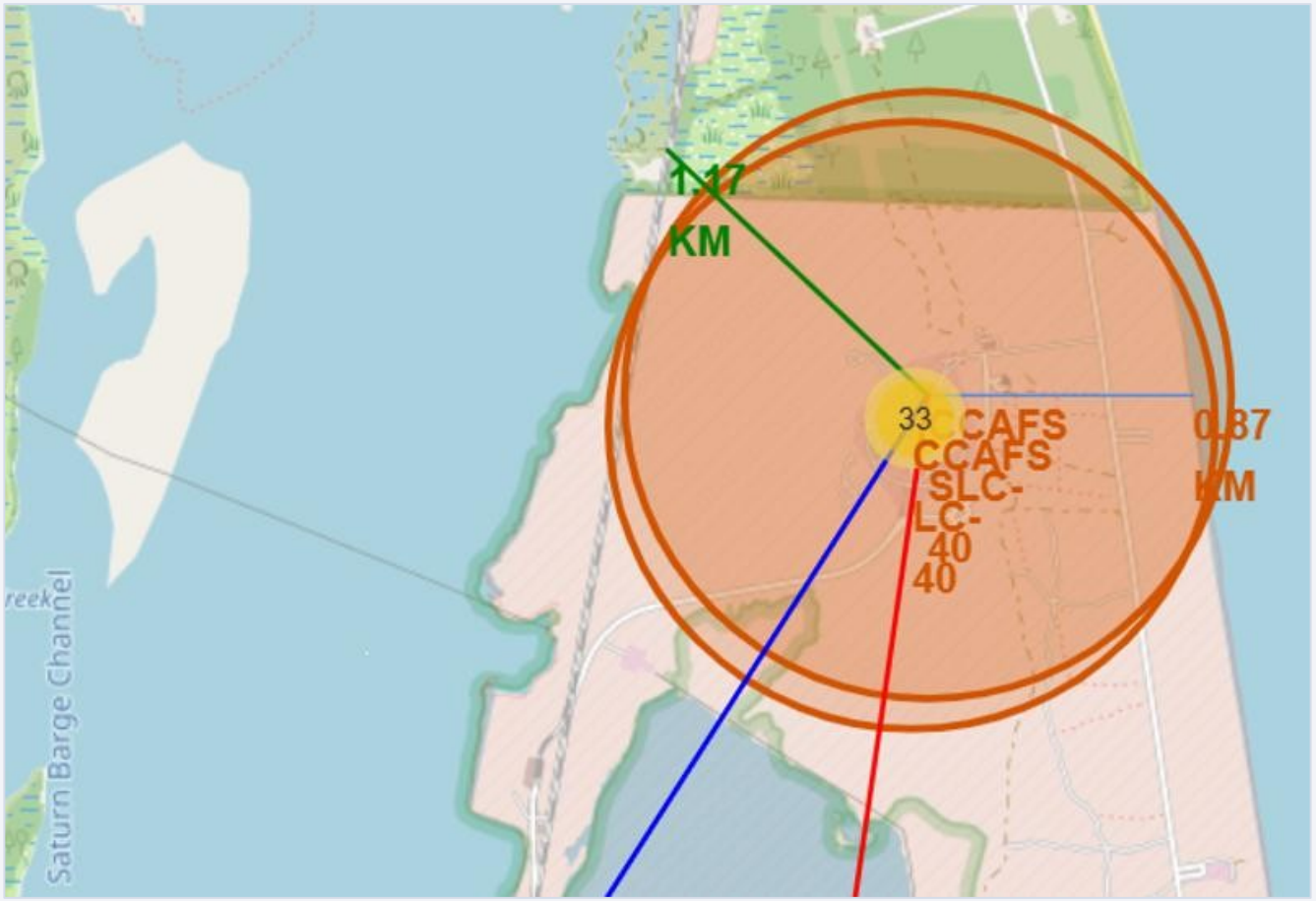




- Launches are grouped in clusters per site; expanded, each marker is one launch: green = landed, red = did not land.
- **Top – KSC LC-39A:** 10 of 13 green (77%), the best ratio.
- **Bottom – CCAFS LC-40 + SLC-40 (shared cluster):** 10 of 33 green; the early, mostly failed flights flew from here.
- VAFB SLC-4E: 4 of 10 landed (40%).

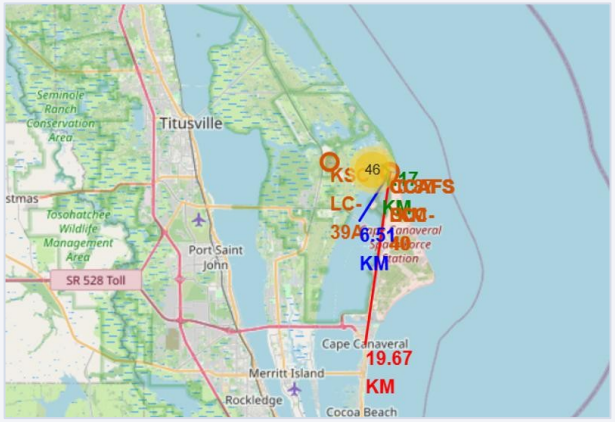
Map data © OpenStreetMap contributors

CCAFS SLC-40 and Its Proximities



- **Coastline 0.87 km** – Atlantic beside the pad
- **Railway 1.17 km** – NASA Railroad (plus the pad's own spur)
- **Highway 6.51 km** – NASA Parkway East
- **City 19.67 km** – Cape Canaveral town

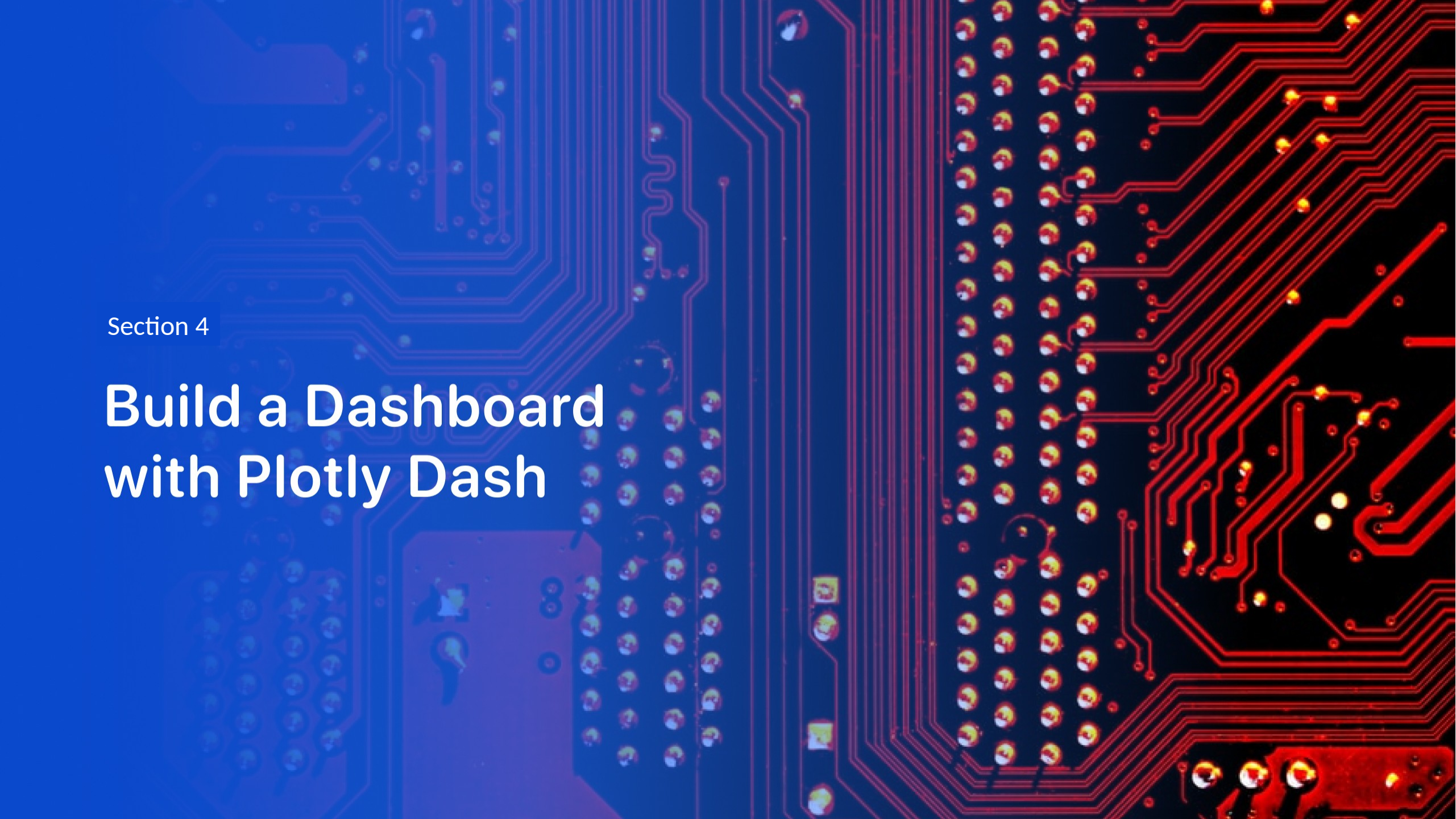
Close to the sea and to rail, but kept well away from people.



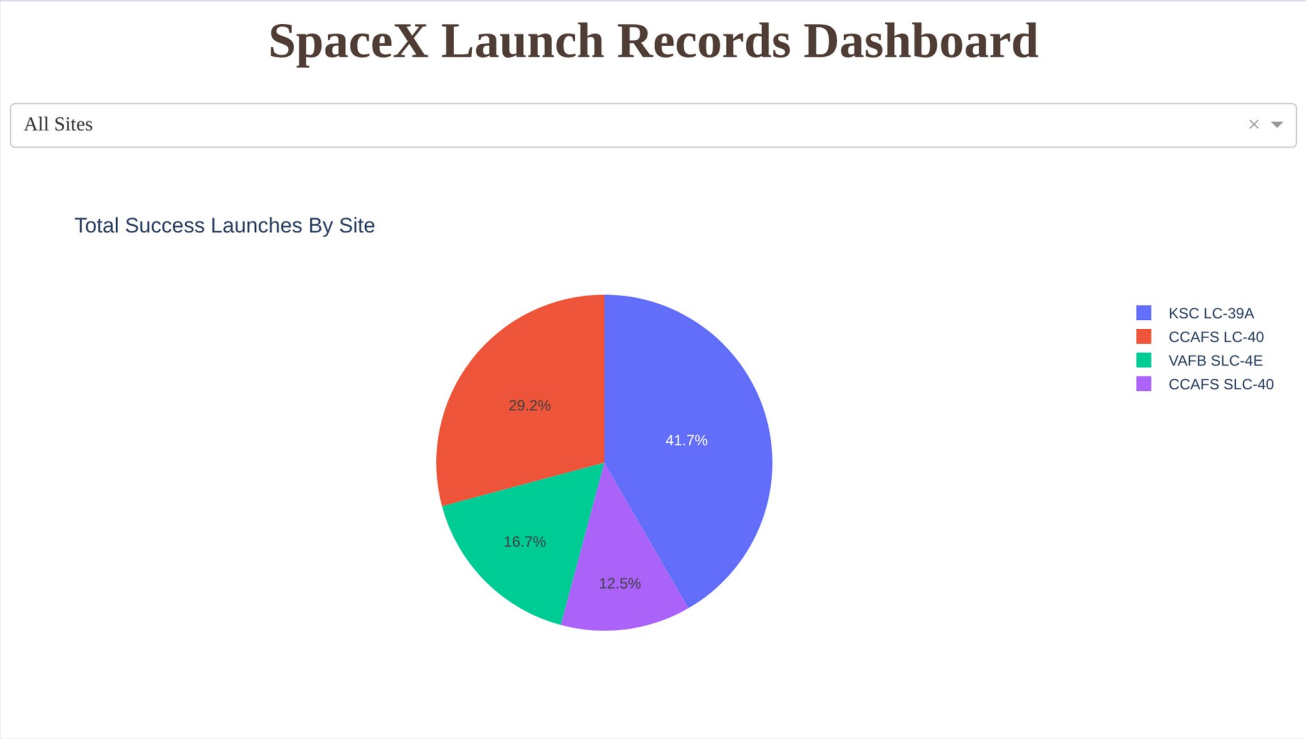
Left: pad close-up. Right: lines to highway and town. Map data © OpenStreetMap contributors

Section 4

Build a Dashboard with Plotly Dash

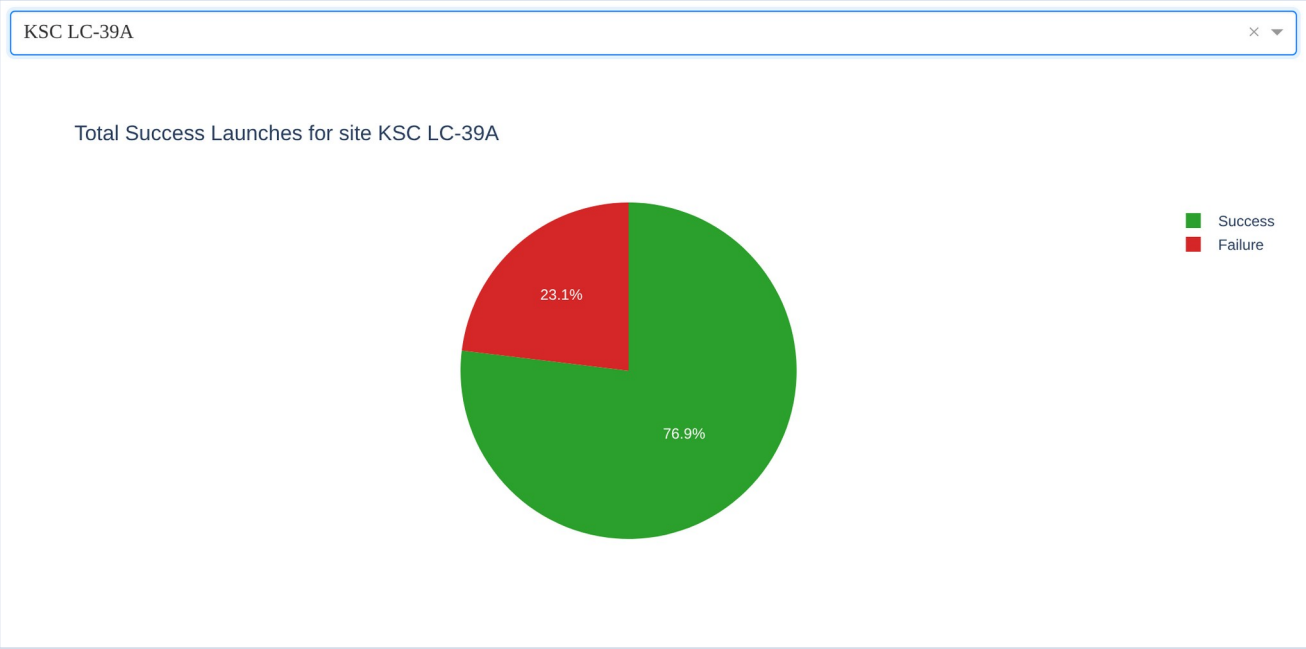


Total Successful Launches by Site



- With 'All Sites' selected, the pie shows each site's share of all successful landings.
- KSC LC-39A has the most successes (41.7%, 10 landings), followed by CCAFS LC-40 (29.2%), VAFB SLC-4E (16.7%) and CCAFS SLC-40 (12.5%).
- 24 successful landings in total in the dashboard data (56 launches).

KSC LC-39A: Highest Launch Success Ratio

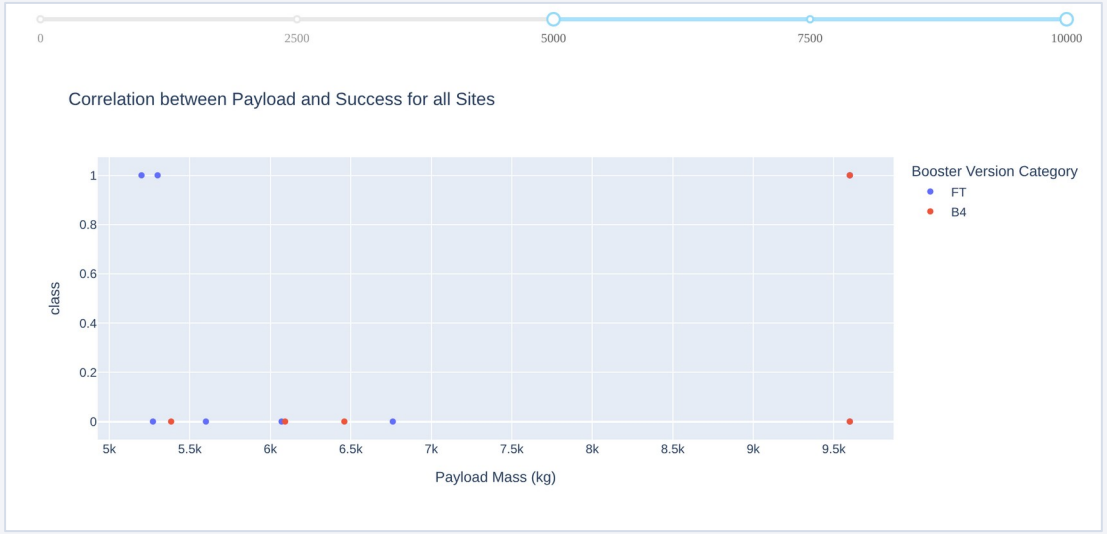


- Selecting KSC LC-39A shows its own success vs. failure split.
- 76.9% of its launches landed (10 of 13) — the best ratio of any site.
- For comparison: CCAFS SLC-40 43%, VAFB SLC-4E 40%, CCAFS LC-40 27%.

Payload vs. Launch Outcome for All Sites



Slider 0 – 5,000 kg



Slider 5,000 – 10,000 kg

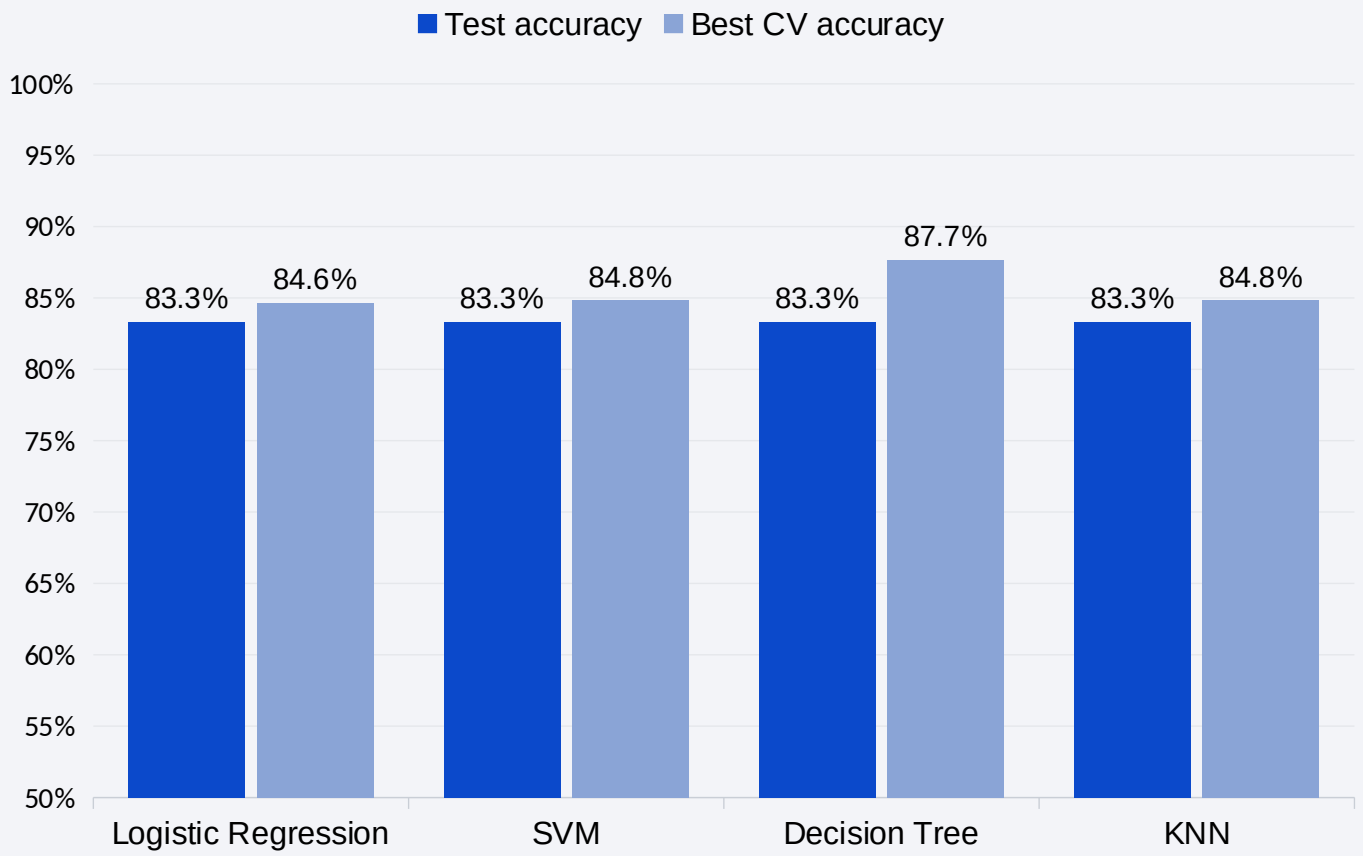
- 2,000–4,000 kg has the highest success rate (62%); 6,000–8,000 kg the lowest (0 of 4), with 0–2,000 kg also weak (23%).
- By booster version, FT is the best version with a real sample (67% of 24); v1.0 (0 of 5) and v1.1 (1 of 15) rarely landed. B5 is 1 of 1.
- Light and heavy payloads both fail with old boosters — booster version matters more than payload.

Section 5

Predictive Analysis (Classification)

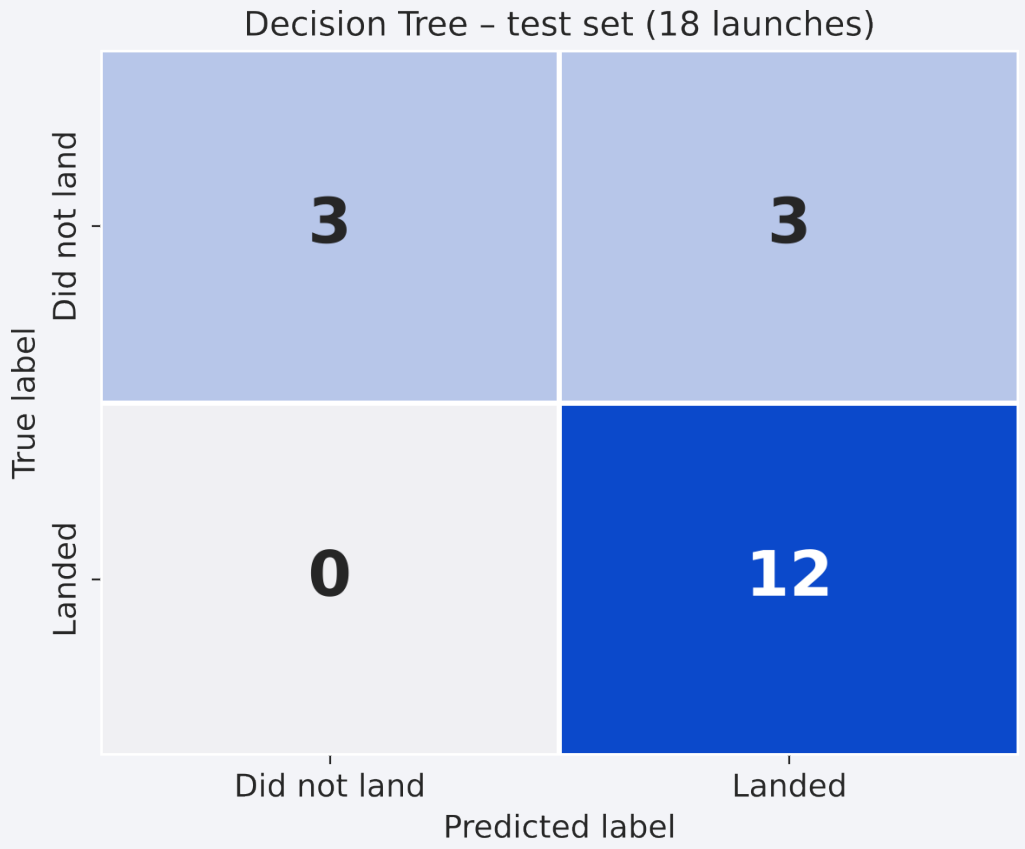


Classification Accuracy



- All four tuned models reach the same 83.3% accuracy on the 18-launch test set (15 correct).
- The tie is broken by cross-validation: the Decision Tree scores 87.7%, ahead of SVM and KNN (84.8%) and Logistic Regression (84.6%).
- **Best model: Decision Tree**
  - gini, max_depth 8, max_features sqrt, min_samples_leaf 2, min_samples_split 10

Confusion Matrix



Decision Tree on the 18 test launches:

- **12 true positives** — every landing was predicted correctly.
- **3 true negatives** — failed landings correctly flagged.
- **3 false positives** — failed landings predicted as landed.
- **0 false negatives.**

The model is optimistic: it never misses a landing but over-predicts success for half of the failures. The other three models make exactly the same errors.

Conclusions

Experience pays off Landing success rose from 0% before 2014 to 84–90% in 2019–2020; later flights land far more often.

Site matters KSC LC-39A has the most successful landings and the best ratio (77%).

Orbit and payload matter SSO, VLEO and LEO missions land most often; GTO is the hardest. Heavy payloads to LEO/Polar/ISS land well.

Booster version matters Full Thrust and Block 5 boosters land; v1.0 and v1.1 rarely did.

Location All sites are coastal, close to rail, and ≥ 20 km from towns.

Prediction A tuned Decision Tree predicts landings with 83% test accuracy (88% CV) — good enough to support Space Y's bid pricing, though it over-predicts success.

Appendix

File (GitHub)	Content
jupyter-labs-spacex-data-collection-api-v2.ipynb	Data collection – SpaceX API
jupyter-labs-webscraping.ipynb	Data collection – web scraping
labs-jupyter-spacex-Data wrangling-v2.ipynb	Data wrangling
jupyter-labs-eda-sql-coursera_sqllite.ipynb	EDA with SQL
jupyter-labs-eda-dataviz-v2.ipynb	EDA with visualization
lab-jupyter-launch-site-location-v2.ipynb	Interactive map with Folium
spacex_dash_app.py	Plotly Dash dashboard
SpaceX-Machine-Learning-Prediction-Part-5-v1.ipynb	Machine learning prediction

Execution notes

- The public SpaceX API (api.spacexdata.com) was offline (HTTP 522) in October 2026, so the API notebook's per-ID lookups were served from IBM's archived copies of the same API responses; the output reproduces IBM's dataset_part_1.csv exactly.
- Scikit-learn 1.3+ removed max_features='auto'; the Decision Tree grid uses 'sqrt' and 'log2' instead.
- Repository: https://github.com/marcosroquerosa/ibm-data-science-capstone

Thank you!

